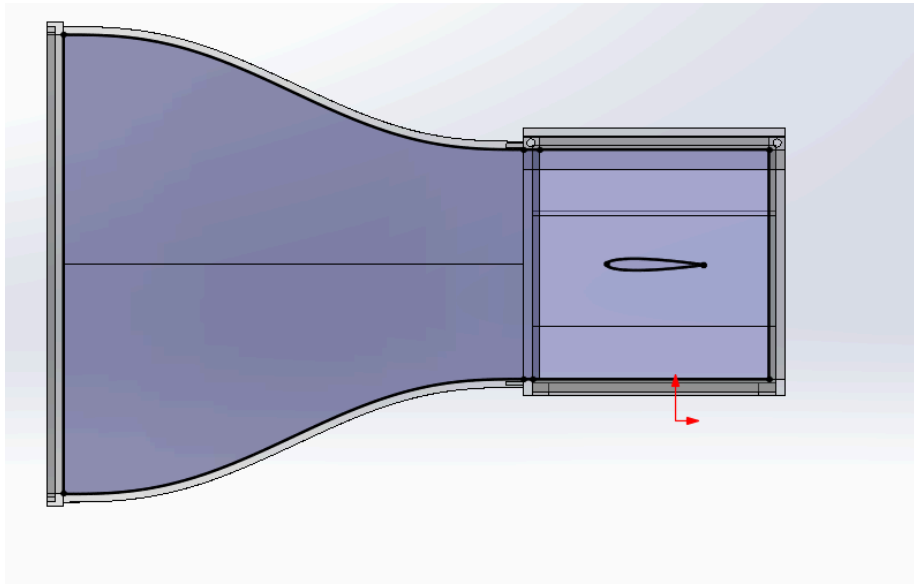
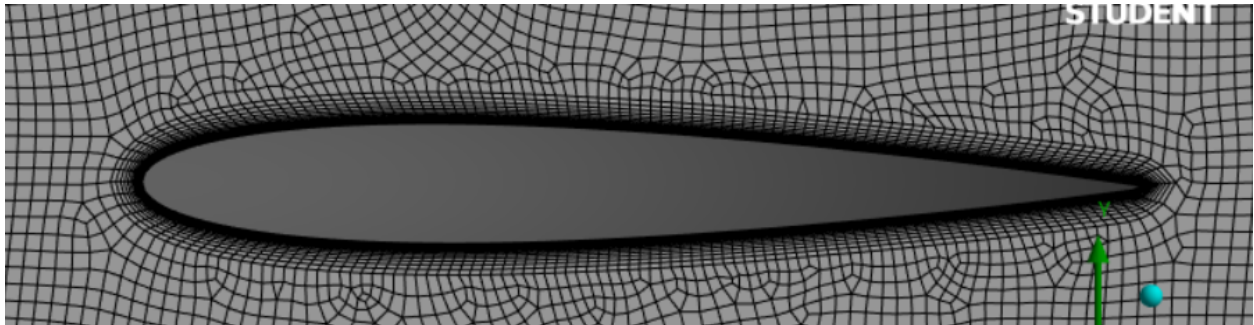


NACA 0012 Ansys Resultant Data

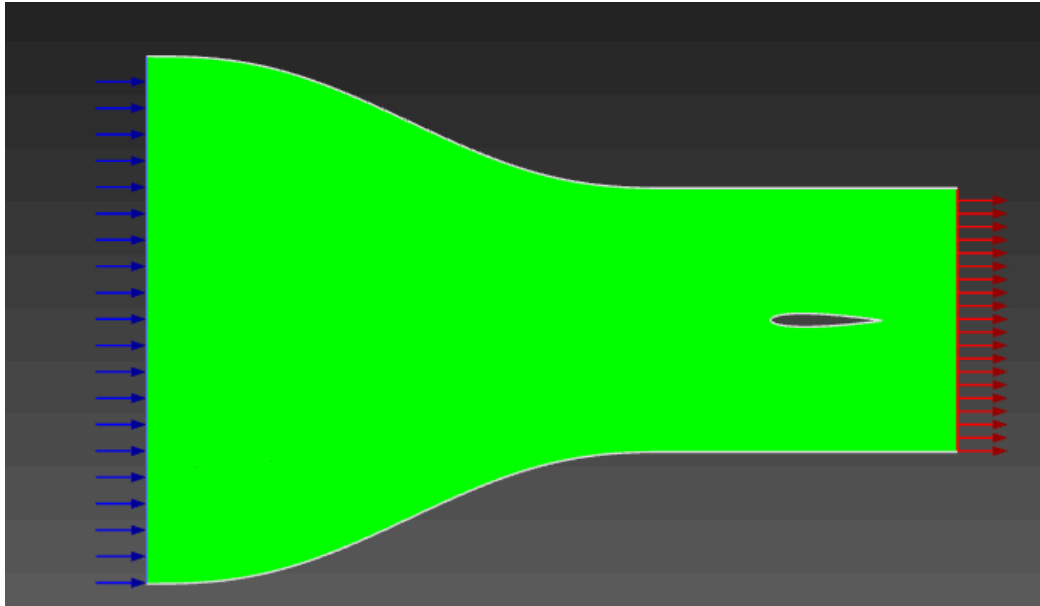
Attached is all information and data as calculated for the NACA 0012 Airfoil. Pictured below is the cad model of both the contraction chamber and test chamber. In dark black the 2d planar surface used for analysis is depicted. Rather than utilizing 3d airfoil analysis which is limited on the student version, 2d analysis was done. This allows for in depth meshing and high quality flow profiles.



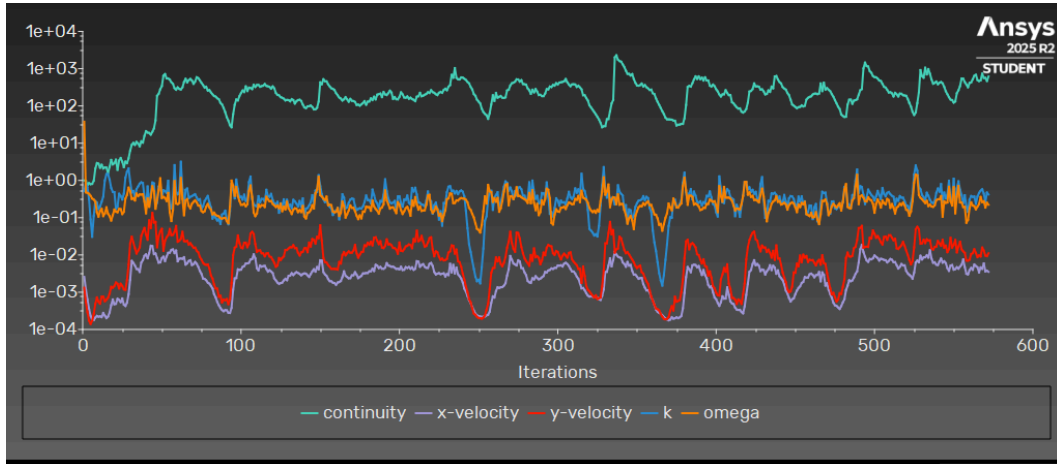
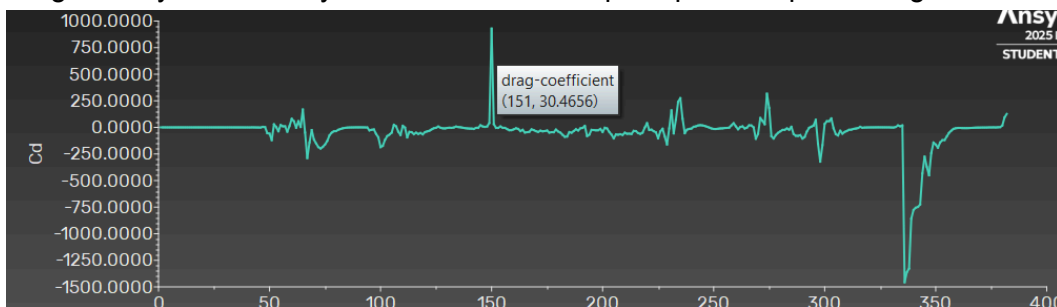
Contraction chamber is 14x14", whereas the test chamber is 7x7". As mentioned, depth is not depicted so only height is used to calculate. The Naca 0012 at 0 degrees AoA is shown.



There is a decently dense mesh applied around the airfoil. For reference, the airfoil is 3 inches in length. This mesh includes 20 boundary layers with 0.0005 inch initial cell size and 1.2 growth factor. No trailing edge interactions are considered.



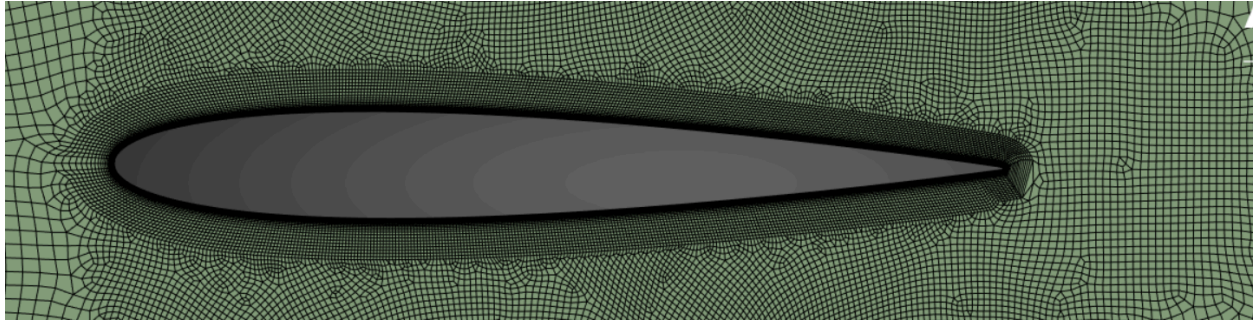
Here is the model in fluent, which describes the velocity vectors. The resolution is too low but the green layer is actually a bunch of microscopic squares representing the mesh.



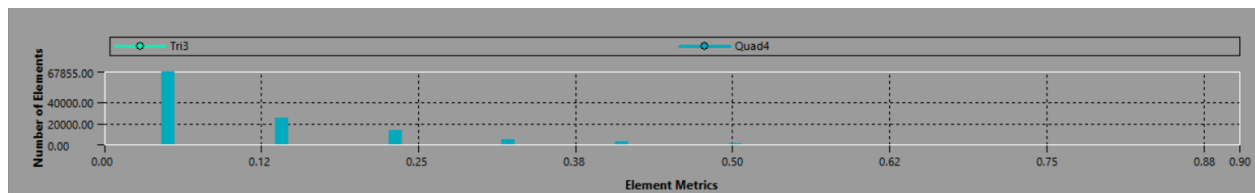
First Attempt: One of the goals of CFD is to design a mass threshold upon which we are satisfied. By the continuity equation, mass in must equal mass out. Because the system is calculating over 1×10^5 equations at once, mass in won't exactly equal mass out. Therefore, we design thresholds on residuals, or the amount of error we're okay with. The error in the case is 10^{-6} for all parameters. As depicted in the second image, the first attempt did not yield the expected data trends. Instead, the data was scattered. This meant that the solver was getting

stuck, recalculating but not able to reduce error. This is demonstrated in the coefficient of drag curve, where results are still wildly fluctuating after 300 iterations. By then, we expect the curve to have stabilized to a value. This resulted from a poor boundary layer that was under defined, explicitly resulting in a singularity at the suction peak. This is not depicted, but velocity reached 5×10^4 indicating a singularity.

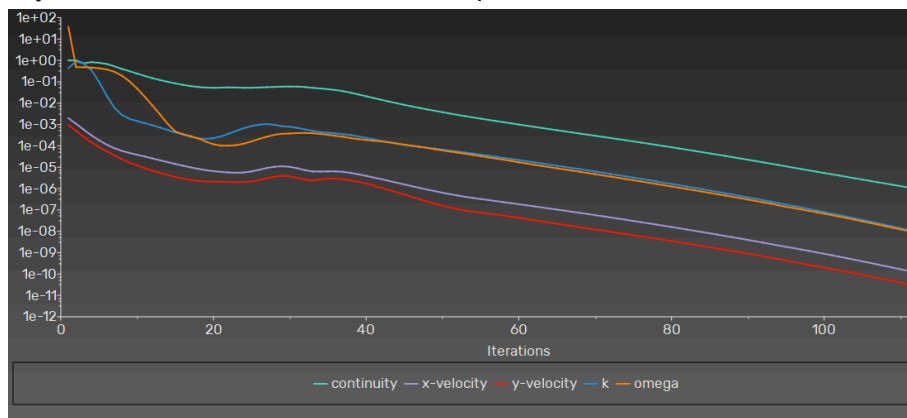
Attempt 2

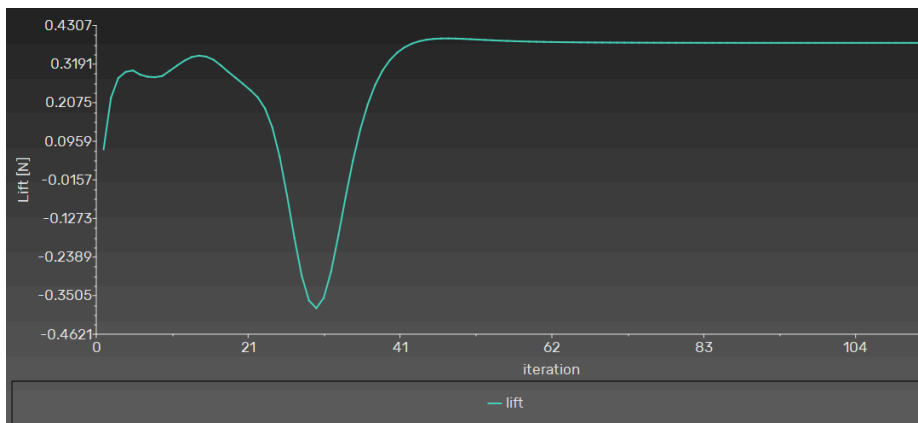
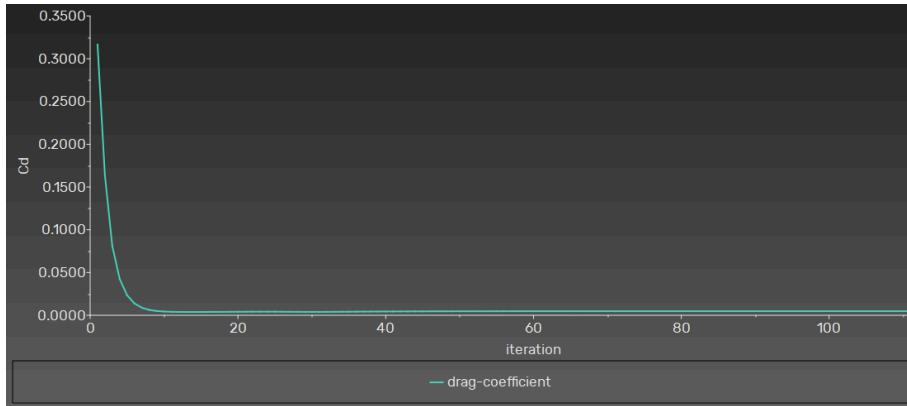


Here is the refined mesh, which has a much higher cell density than previous, and more accuracy around the airfoil. There's a 3.6 inch sphere of influence with element size 0.02 around the airfoil. Not shown is the rest of the mesh which contains 0.05 inch cell sizes. A boundary layer is also attached so as to more accurately model turbulent conditions around the airfoil. This uses a growth rate of 1.2, with 30 layers and 0.0002 inch size at the smallest layer. Finally, a similar sphere of influence is attached at the end to account for trailing edge interactions.

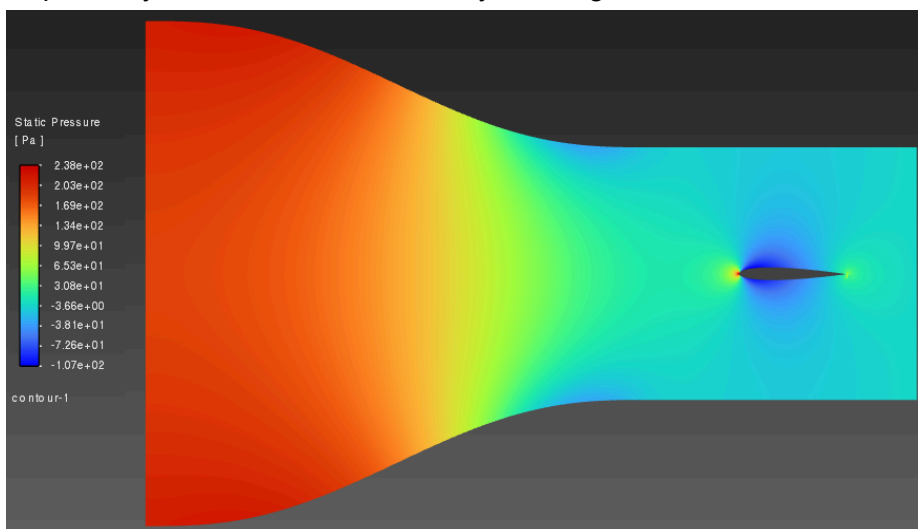


The above table shows skewness, or cell quality. The goal is to have as many cells near 0 as possible. As skewness increases, data quality decreases because the cells are more asymmetric and the mathematical equations used don't work as well.



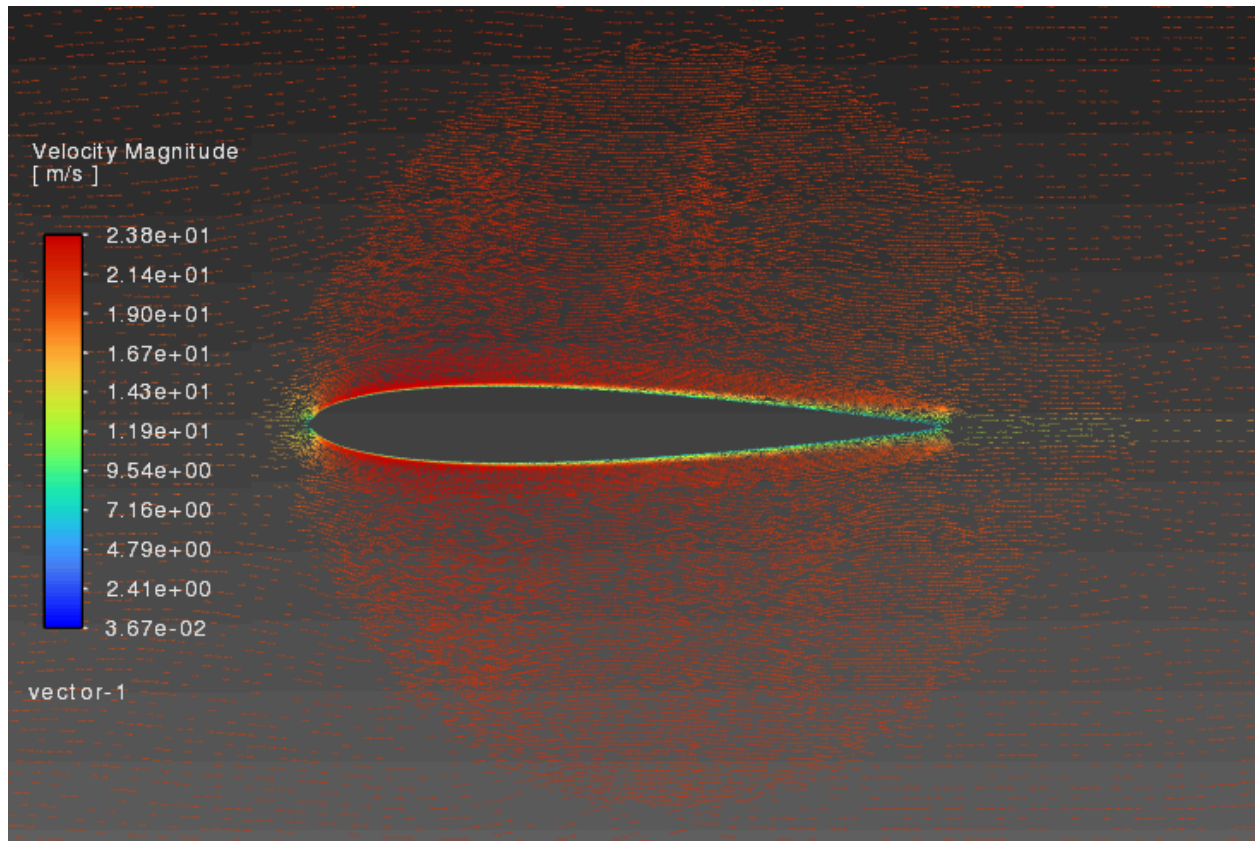


The residuals shown in the first picture match exactly what we want. The trend downwards logarithmically, and in fact, the data was so high quality that only 110 iterations were needed to satisfy all parameters and converge on values. The convergence process can be better visualized in lift and drag coefficient data, which settled on 0.38 newtons and 0.0048 respectively. Coefficient of lift similarly converged to 0.005

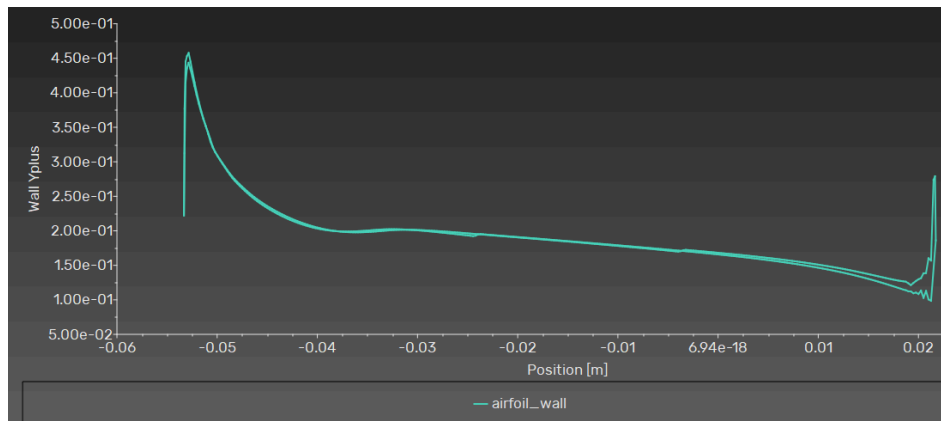


Here is a really useful plot of static pressure. By bernoulli's principle,

$P_1 + 1/2\rho v_1^2 = P_2 + 1/2\rho v_2^2$, As velocity increases, we need pressure to decrease which is displayed above. More importantly are the interactions at the airfoil surface. Firstly, there's a pressure spike at the very edge of the airfoil, which occurs due to a stagnation point where velocity drops to 0. Secondly, there's a suction peak on both the top and the bottom, which occurs due to airfoil symmetry. The suction peak occurs because of the way velocity is driven over the curve, creating a pressure gradient at that point. Also visualized are the interactions at the walls, showing how they have an impact on the flow.



This is a vector plot showing the magnitude of velocity at points along the curve. At the points where velocity is driven up, the suction peak occurs.



Y+ value is also low, meaning that the physics is calculated directly.

